Lab 6: Code and Branch Coverage

Repository: https://github.com/sanchita-suni/L6_L7_54_museum_admission

Code Coverage

coverage run -m pytest CodeCoverageTest_54_museum_admission.py

```
/workspace/L6_L7_54_museum_admission/LAB6$ coverage run -m pytest CodeCoverageTest_54_museum_admission.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /workspace/L6_L7_54_museum_admission/LAB6
plugins: platformdirs-4.13.3, anyio-4.15.1
collected 14 items

CodeCoverageTest_54_museum_admission.py ..............                   [100%]

============================== 14 passed in 0.05s ==============================
/workspace/L6_L7_54_museum_admission/LAB6$ coverage report -m
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
CodeCoverageTest_54_museum_admission.py      33      0   100%
museum_admission.py                          36      0   100%
---------------------------------------------------------------------
TOTAL                                        69      0   100%
/workspace/L6_L7_54_museum_admission/LAB6$
```

coverage run --branch -m pytest CodeCoverageTest_54_museum_admission.py

```
/workspace/L6_L7_54_museum_admission/LAB6$ coverage run --branch -m pytest CodeCoverageTest_54_museum_admission.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /workspace/L6_L7_54_museum_admission/LAB6
plugins: platformdirs-4.13.3, anyio-4.15.1
collected 14 items

CodeCoverageTest_54_museum_admission.py ..............                   [100%]

============================== 14 passed in 0.03s ==============================
/workspace/L6_L7_54_museum_admission/LAB6$ coverage report -m
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
CodeCoverageTest_54_museum_admission.py      33      0      0      0   100%
museum_admission.py                          36      0     28      2    97%   14->16, 16->21
-----------------------------------------------------------------------------------
TOTAL                                        69      0     28      2    98%
/workspace/L6_L7_54_museum_admission/LAB6$
```

Branch Coverage

coverage run -m pytest BranchCoverageTest_54_museum_admission.py

```
/workspace/L6_L7_54_museum_admission/LAB6$ coverage run -m pytest BranchCoverageTest_54_museum_admission.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /workspace/L6_L7_54_museum_admission/LAB6
plugins: platformdirs-4.13.3, anyio-4.15.1
collected 36 items

BranchCoverageTest_54_museum_admission.py ............................ [ 83%]
......                                                                 [100%]

============================== 36 passed in 0.13s ==============================
/workspace/L6_L7_54_museum_admission/LAB6$ coverage report -m
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
BranchCoverageTest_54_museum_admission.py      57      0   100%
museum_admission.py                            36      0   100%
---------------------------------------------------------------------
TOTAL                                          93      0   100%
/workspace/L6_L7_54_museum_admission/LAB6$
```

coverage run --branch -m pytest BranchCoverageTest_54_museum_admission.py

```
/workspace/L6_L7_54_museum_admission/LAB6$ coverage run --branch -m pytest BranchCoverageTest_54_museum_admission.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /workspace/L6_L7_54_museum_admission/LAB6
plugins: platformdirs-4.13.3, anyio-4.15.1
collected 36 items

BranchCoverageTest_54_museum_admission.py ............................ [ 83%]
......                                                                 [100%]

============================== 36 passed in 0.06s ==============================
/workspace/L6_L7_54_museum_admission/LAB6$ coverage report -m
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
BranchCoverageTest_54_museum_admission.py      57      0      0      0   100%
museum_admission.py                            36      0     28      0   100%
-----------------------------------------------------------------------------------
TOTAL                                          93      0     28      0   100%
/workspace/L6_L7_54_museum_admission/LAB6$
```

CodeCoverageTest_54_museum_admission.py

```
"""Code (line) coverage tests for museum_admission.py.

Goal: execute every statement in the module at least once.
Run:
    coverage run -m pytest CodeCoverageTest_54_museum_admission.py
    coverage report -m
"""
import pytest
from museum_admission import ticket_price, group_discount, gallery_entry


# ---------- ticket_price ----------

def test_unknown_category_raises():
    with pytest.raises(ValueError, match="unknown category"):
        ticket_price("vip", 30, "tuesday", False)


def test_negative_age_raises():
    with pytest.raises(ValueError, match="invalid age"):
        ticket_price("adult", -1, "tuesday", False)


def test_child_under_five_is_free():
    assert ticket_price("child", 3, "saturday", True) == 0


def test_student_monday_with_guided_tour():
    # 150 halved on Monday = 75, student tour adds 50
    assert ticket_price("student", 20, "monday", True) == 125


def test_senior_sunday_with_guided_tour():
    # 200 + 50 weekend surcharge, non-student tour adds 100
    assert ticket_price("senior", 65, "sunday", True) == 350


# ---------- group_discount ----------

def test_group_size_zero_raises():
    with pytest.raises(ValueError, match="invalid group size"):
        group_discount(0, False)


def test_large_school_group():
    assert group_discount(25, True) == 40


def test_small_school_group():
    assert group_discount(10, True) == 25


def test_large_public_group():
    assert group_discount(15, False) == 20


def test_medium_public_group():
    assert group_discount(8, False) == 10
```

```
def test_small_public_group():
    assert group_discount(4, False) == 0


# ---------- gallery_entry ----------

def test_gallery_full_must_wait():
    assert gallery_entry(100, 100, False) == "wait"


def test_gallery_nearly_full_without_slot_queues():
    assert gallery_entry(90, 100, False) == "queue"


def test_gallery_with_space_enters():
    assert gallery_entry(10, 100, False) == "enter"
```

BranchCoverageTest_54_museum_admission.py

```
"""Branch coverage tests for museum_admission.py.

Goal: every decision (if / elif / else, and/not conditions) evaluates to
both True and False at least once, plus the boundary values of each rule.
Run:
    coverage run --branch -m pytest BranchCoverageTest_54_museum_admission.py
    coverage report -m
"""
import pytest
from museum_admission import ticket_price, group_discount, gallery_entry


# ---------- ticket_price ----------

def test_unknown_category_raises():
    with pytest.raises(ValueError, match="unknown category"):
        ticket_price("vip", 30, "tuesday", False)


def test_negative_age_raises():
    with pytest.raises(ValueError, match="invalid age"):
        ticket_price("adult", -1, "tuesday", False)


def test_age_zero_is_free():
    assert ticket_price("adult", 0, "tuesday", False) == 0


def test_age_four_is_free():
    assert ticket_price("child", 4, "saturday", True) == 0


def test_age_five_pays():
    # boundary: age < 5 is False at exactly 5
    assert ticket_price("child", 5, "tuesday", False) == 100


@pytest.mark.parametrize(
    "category, expected",
    [("adult", 300), ("student", 150), ("child", 100), ("senior", 200)],
)
def test_weekday_without_tour_is_base_price(category, expected):
    # day is neither Monday nor weekend, no guided tour: both False branches
    assert ticket_price(category, 30, "wednesday", False) == expected


def test_monday_halves_price_without_tour():
    assert ticket_price("adult", 30, "monday", False) == 150


def test_monday_odd_price_uses_floor_division():
    assert ticket_price("student", 20, "monday", False) == 75


def test_saturday_surcharge_without_tour():
    assert ticket_price("adult", 30, "saturday", False) == 350


def test_sunday_surcharge_without_tour():
    assert ticket_price("child", 8, "sunday", False) == 150
```

```
def test_student_guided_tour_adds_fifty():
    assert ticket_price("student", 20, "tuesday", True) == 200


def test_non_student_guided_tour_adds_hundred():
    assert ticket_price("adult", 30, "tuesday", True) == 400


def test_student_monday_with_guided_tour():
    assert ticket_price("student", 20, "monday", True) == 125


def test_senior_sunday_with_guided_tour():
    assert ticket_price("senior", 65, "sunday", True) == 350


def test_day_is_case_sensitive():
    # "Monday" is not "monday", so no discount is applied
    assert ticket_price("adult", 30, "Monday", False) == 300


# ---------- group_discount ----------

@pytest.mark.parametrize("size", [0, -3])
def test_invalid_group_size_raises(size):
    with pytest.raises(ValueError, match="invalid group size"):
        group_discount(size, False)


@pytest.mark.parametrize(
    "size, expected",
    [(1, 25), (19, 25), (20, 40), (45, 40)],
)
def test_school_groups(size, expected):
    # is_school True, with both sides of the size >= 20 boundary
    assert group_discount(size, True) == expected


@pytest.mark.parametrize(
    "size, expected",
    [(1, 0), (7, 0), (8, 10), (14, 10), (15, 20), (30, 20)],
)
def test_public_groups(size, expected):
    # is_school False, with both sides of the 8 and 15 boundaries
    assert group_discount(size, False) == expected


# ---------- gallery_entry ----------

def test_gallery_at_capacity_waits():
    assert gallery_entry(100, 100, True) == "wait"


def test_gallery_over_capacity_waits():
    assert gallery_entry(120, 100, False) == "wait"


def test_gallery_at_ninety_percent_without_slot_queues():
    # boundary: 90 * 10 >= 100 * 9
    assert gallery_entry(90, 100, False) == "queue"
```

```
def test_gallery_at_ninety_percent_with_slot_enters():
    # first condition True, has_timed_slot makes the whole check False
    assert gallery_entry(90, 100, True) == "enter"


def test_gallery_just_below_ninety_percent_enters():
    # first condition False even without a timed slot
    assert gallery_entry(89, 100, False) == "enter"


def test_empty_gallery_enters():
    assert gallery_entry(0, 50, False) == "enter"
```